Week 5: Commitment and Strategic Moves

Credibility, Irreversibility, and Changing Future Incentives

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A commitment is powerful only when it changes what will be rational later.

From Week 4 to Week 5

Week 4: Preserve Flexibility

- Randomise over actions
- Avoid predictable patterns
- Keep the rival indifferent
- Reduce exploitation

Week 5: Restrict Flexibility

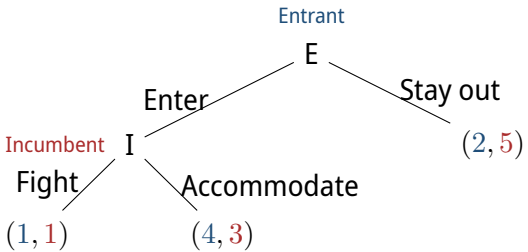
- Act before the rival
- Make reversal costly
- Change future incentives
- Influence the rival's response

Today, an action taken now changes how the future game is played.

Opening Decision: Can a Threat Deter Entry?

The Situation

An entrant chooses **Enter** or **Stay out**. After entry, the incumbent chooses **Fight** or **Accommodate**.



Does saying "I will fight" change the outcome?

First payoff: entrant

Second payoff: incumbent

Today's Learning Outcomes

By the end of class, you should be able to:

1. distinguish commitment from cheap talk or ordinary expenditure;
2. test whether a threat or promise is credible;
3. explain why observability and reversibility matter;
4. distinguish sunk cost from a change in future incentives;
5. solve a game before and after commitment;
6. recommend commitment only when its benefit exceeds its cost.

What Is Strategic Commitment?

Definition

A strategic commitment is an observable earlier action that changes a player's future incentives or feasible choices and therefore changes how others respond.

- It occurs before the rival's relevant response.
- The rival can observe or reliably infer it.
- Reversal is sufficiently costly or constrained.
- It changes future payoffs, information, constraints, or actions.

Commitment changes the game; cheap talk describes the game you wish existed.

Credibility: Test the Future Decision

Credible

The promised or threatened action will be optimal, or unavoidable, when the future decision point is reached.

Not Credible

The player will prefer to abandon the announcement after the other side has acted.

- Test the action at the future node.
- Use backward induction, not confidence or rhetoric.
- Threats and promises face the same time-consistency problem.

Costly Signal or Strategic Commitment?

Costly Signal

Changes what others believe about your type, resources, or willingness.

Example: visible spending may signal determination.

Commitment Device

Changes future payoffs, constraints, information, or available actions.

Example: a contract adds a future penalty.

Key Distinction

Cost can communicate seriousness, but spending alone does not necessarily make a future action rational.

Sunk Costs: Important but Often Misunderstood

Sunk Cost

A cost already incurred and unrecoverable at the time of a later decision.

Decision Rule

Ignore a common past cost when choosing later. Ask whether the earlier action changed future revenues, costs, constraints, information, or options.

- “We spent too much to stop” can be a sunk-cost fallacy.
- A factory matters if it changes capacity or marginal cost.
- A contract matters if withdrawal triggers a future penalty.
- The mechanism is the changed continuation game, not the historical expense.

Four Commitment Devices

Device	How It Changes the Game	Illustration
Capacity	Changes future capability or marginal cost	Factory, computing power
Contract	Adds future rewards, duties, or penalties	Supply agreement, guarantee
Delegation	Transfers control or removes discretion	Pricing mandate, automatic trigger
Architecture	Makes reversal or incompatibility costly	Dedicated system, smart contract

The best device is the least costly reliable change to the future game.

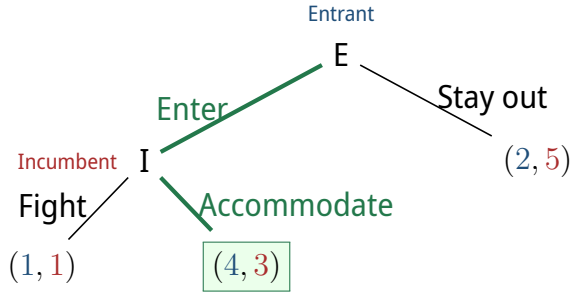
How to Analyse a Commitment

1. Solve the game **without** commitment.
2. Specify the earlier commitment and its cost.
3. State what future payoff, constraint, information, or action changes.
4. Confirm that the relevant player observes the change.
5. Solve the altered game by backward induction.
6. Compare net payoffs with commitment, without commitment, and under alternatives.

Important

A commitment can be credible but still unprofitable.

Without Commitment: Entry Is Accommodated



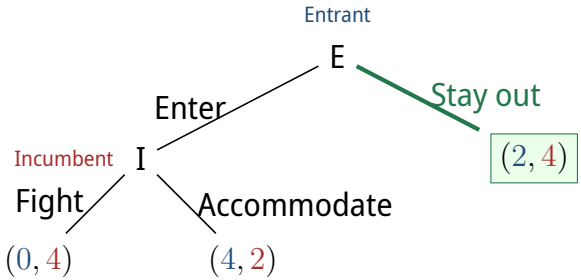
- The incumbent accommodates because $3 > 1$.
- The entrant therefore enters because $4 > 2$.

Predicted path: Enter, then Accommodate.

With Commitment: Future Incentives Change

Illustrative Commitment

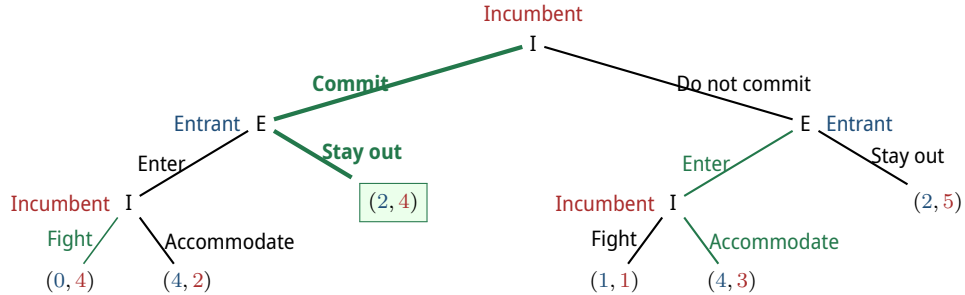
The incumbent pays 1 for a visible supply contract. The contract expands capacity, raises the return from fighting, and reduces the value of accommodating entry.



- After entry, the incumbent now fights because $4 > 2$.
- Anticipating Fight, the entrant stays out because $2 > 0$.

The displayed incumbent payoffs are net of the commitment cost of 1.

The Complete Game: Should the Incumbent Commit?



Commit branch: Fight is credible, so the entrant stays out.
Incumbent payoff: 4, net of cost.

Do-not-commit branch: The incumbent accommodates, so the entrant enters.
Incumbent payoff: 3.

Because $4 > 3$, the incumbent commits.

First payoff: entrant

Second payoff: incumbent

Commitment Quality Checklist

- **Prior:** Is the move made before the relevant response?
- **Observable:** Can the other player verify it?
- **Difficult to reverse:** Is withdrawal costly or constrained?
- **Payoff relevant:** Does it change future incentives or actions?
- **Hard to imitate:** Can the rival copy or bypass it quickly?
- **Proportionate:** Does the strategic benefit exceed the cost?
- **Robust:** Does the conclusion survive reasonable assumption changes?

Application 1: Capacity or Supplier Commitment

Group Task

Select a real firm considering early capacity, dedicated infrastructure, supplier exclusivity, or another observable commitment.

Build the story:

- What is the baseline game?
- What is the commitment action?
- What later payoff or option changes?

Test the recommendation:

- Can the rival observe and imitate it?
- How does the rival respond?
- Is the benefit greater than the cost?

Application 2: When a Contract Backfires

Illustrative Problem

A platform signs a long-term content or technology contract. The contract creates credibility, but it may also reduce quality incentives or lock the firm into an unattractive relationship.

Possible Benefit

- Secures scarce content or technology
- Aligns investment incentives
- Limits future withdrawal

Possible Cost

- Supplier reduces effort or quality
- Buyer loses bargaining flexibility
- Technology or demand changes

A commitment can work exactly as written and still create the wrong incentives.

Commitment Versus Flexibility

Commitment Is More Valuable When...	Flexibility Is More Valuable When...
rival response is sensitive to future incentives;	demand, technology, or regulation is uncertain;
scarce resources can be secured early;	assets can become obsolete quickly;
reversal would destroy credibility;	staged learning is valuable;
the rival cannot imitate easily;	the rival can copy or bypass cheaply.

Core Trade-off

Commitment can improve strategic influence while reducing the ability to adapt.

Application Debrief

1. What future payoff, information, action, or constraint changed?
2. Could the rival observe, verify, imitate, or bypass the commitment?
3. Did backward induction actually change the predicted response?
4. Was the commitment useful, ineffective, or harmful?
5. Could a staged or contractual alternative preserve more flexibility?

Managerial Lesson

The best commitment is not the largest expenditure; it is the least costly reliable change to the continuation game.

Optional Extension: The AI Infrastructure Decision

Save for the Start of Next Class

A Hong Kong technology startup is considering dedicated AI infrastructure, a long-term cloud-capacity contract, or flexible on-demand services. A larger rival observes the decision.

1. Build the baseline game and explain the timing.
2. Specify how each infrastructure option changes future cost, capacity, information, or control.
3. Assign and defend the payoffs, including commitment cost.
4. Solve the multi-stage game by backward induction.
5. Recommend Commit, Contract, Stage, Partner, or Wait.

Week 5: Main Takeaways

1. Commitment is an earlier move that changes the future game.
2. Credibility is tested at the future decision node.
3. Observability and difficulty of reversal support commitment.
4. A costly signal and a commitment device are not identical.
5. Sunk expenditure alone does not determine a later action.
6. Capacity, contracts, delegation, and architecture can change incentives.
7. Strategic benefit must exceed commitment and flexibility costs.

Solve the baseline. Change the game. Solve again. Compare net value.

Exit Ticket and Looking Ahead

Exit Ticket

Why is “We have already spent too much to stop” not a valid commitment argument by itself? What additional change could make continued action credible?

Before Week 6

Complete the pre-reading on repeated games. Come ready to ask:

How can future rewards and punishments change today’s behaviour?

Supplementary: Deriving the Entry-Game Payoffs

Without commitment

- Fight, $(1, 1)$: a price or output war hurts both firms.
- Accommodate, $(4, 3)$: both firms operate; the entrant gains more than under conflict.
- Stay out, $(2, 5)$: the entrant keeps its outside option; the incumbent retains the market.

With the supply contract

- Fight, $(0, 4)$: expanded capacity makes aggressive output profitable and entry unattractive.
- Accommodate, $(4, 2)$: unused contracted capacity makes accommodation less valuable.
- Stay out, $(2, 4)$: the incumbent retains the market but still pays the contract cost of 1.

The values are illustrative. The contract matters because it changes branch-specific future profits, not merely because money was spent.

Supplementary: Formal Credibility Check

Threat After Entry

Let $u_I(F)$ be the incumbent's payoff from Fight and $u_I(A)$ its payoff from Accommodate after entry.

- Fight is credible if $u_I(F) \geq u_I(A)$.
- Fight is an empty threat if $u_I(F) < u_I(A)$.
- Commitment works only if it changes this inequality or removes Accommodate.

Credibility is a property of future incentives, not present determination.

Supplementary: Why a Pure Sunk Cost May Not Work

Suppose a prior expenditure K is unrecoverable under both future actions.

$$u_I(F) = R_F - C_F - K,$$
$$u_I(A) = R_A - C_A - K.$$

$$u_I(F) - u_I(A) = (R_F - C_F) - (R_A - C_A).$$

Result

The common sunk cost K cancels. It does not change the later preference between Fight and Accommodate.

Supplementary: A Simple Profit Foundation

Illustrative Components

Profit = market contribution − competition cost − capacity or contract cost.

- **Fight:** include retained demand, lower marginal cost from capacity, and the cost of aggressive output.
- **Accommodate:** include shared demand and any cost of unused contracted capacity.
- **Stay out:** include monopoly contribution minus the commitment cost already paid.
- **Entrant:** include entry cost, expected operating contribution, and its outside option.

Modelling Discipline

State the assumptions first, derive the payoff ranking second, and solve the tree only after the numbers are internally consistent.

Supplementary: Commitment Cost and Profitability

Commit Only If

$$V_{\text{after rival response}} - K \geq V_{\text{without commitment}}.$$

- K is the direct cost of creating the commitment.
- The commitment may change the rival's response and continuation value.
- Compare with staged, reversible, contractual, or partnership alternatives.

Supplementary: Signalling Versus Commitment

	Signalling	Commitment
Main effect	Changes beliefs about type or intention	Changes future incentives or feasible actions
Key requirement	Cost or evidence differs across types	Earlier move is observable and difficult to reverse
Test	Does the observer update beliefs?	Does the continuation game change?

They May Coexist

A visible capacity investment can both signal resources and change future production incentives.

Supplementary: Common Errors

Common Error	Better Check
Large spending creates credibility.	Show the changed future payoff, constraint, or action.
Sunk cost means the firm must continue.	Ignore common past costs in the later decision.
The rival can only Fight or Concede.	Include differentiation, partnership, delay, or imitation.
Successful deterrence is profitable.	Subtract commitment cost and compare alternatives.
Irreversible means literally impossible to reverse.	Ask whether reversal is sufficiently costly or constrained.